AI and Analytics using Zia

Hotel Booking Management System & Predictive Intelligence Laboratory Report

APPLICATION	AI ENGINE	PLATFORM	STATUS
Hotel Booking System	**Zoho Zia AI**	**Zoho Creator**	**Completed / Active**

1. AIM

To design, build, and deploy an intelligent **Hotel Booking Application** integrated with **Zoho Zia AI** to leverage predictive analytics, automated forecasting, operational performance reporting, and guest preference modeling. The system aims to optimize room rate dynamic pricing, forecast occupancy levels, and analyze revenue trends in real time.

2. SOFTWARE & HARDWARE REQUIREMENTS

Component	Specification / Tool
Development Platform	Zoho Creator Low-Code Enterprise Platform
AI & Analytics Engine	Zoho Zia Artificial Intelligence Suite (Demand Forecasting & Analytics)
Modules Built	Dashboard, Requests, Room Inventories, Guest Preferences, Operational Reports, Forecasts
Web Environment	Modern Web Browser (Chrome / Edge / Firefox)
Data Models	Historical Booking Records, Room Base Rates, Competitor Price Benchmarks, Occupancy Logs

3. IMPLEMENTATION PROCEDURE

1. **Data Schema Configuration:** Created fundamental relational tables for Properties (GHR-001, SBL-002, US-003), Room Inventories, Guest Profiles, and Historical Booking Records.
2. **Dashboard & KPI Setup:** Designed an intuitive user interface to display key hospitality metrics such as Total Revenue ($1,890.00), Total Daily Rates ($900.00), Total Bookings (5,580), and Average Occupancy Rate (70.62%).
3. **Zia AI Integration:** Configured Zia Predictive Models to process historical reservation trends and project future demand cycles across Daily, Monthly, and Quarterly timeframes.
4. **Operational Reporting Module:** Built automated report generators to track property-wise booking counts, staffing schedules, and average occupancy across customized date ranges.
5. **Guest Preference Analytics:** Configured classification groupings to segregate guest choices across Amenity, Experience, and Price tiers for personalized guest retention.

4. EXPERIMENTAL RESULTS & VISUAL PROOFS

$1,890.00	5,580	70.62%	82.10%
Total Amount	Total Bookings	Avg Occupancy	Peak Demand

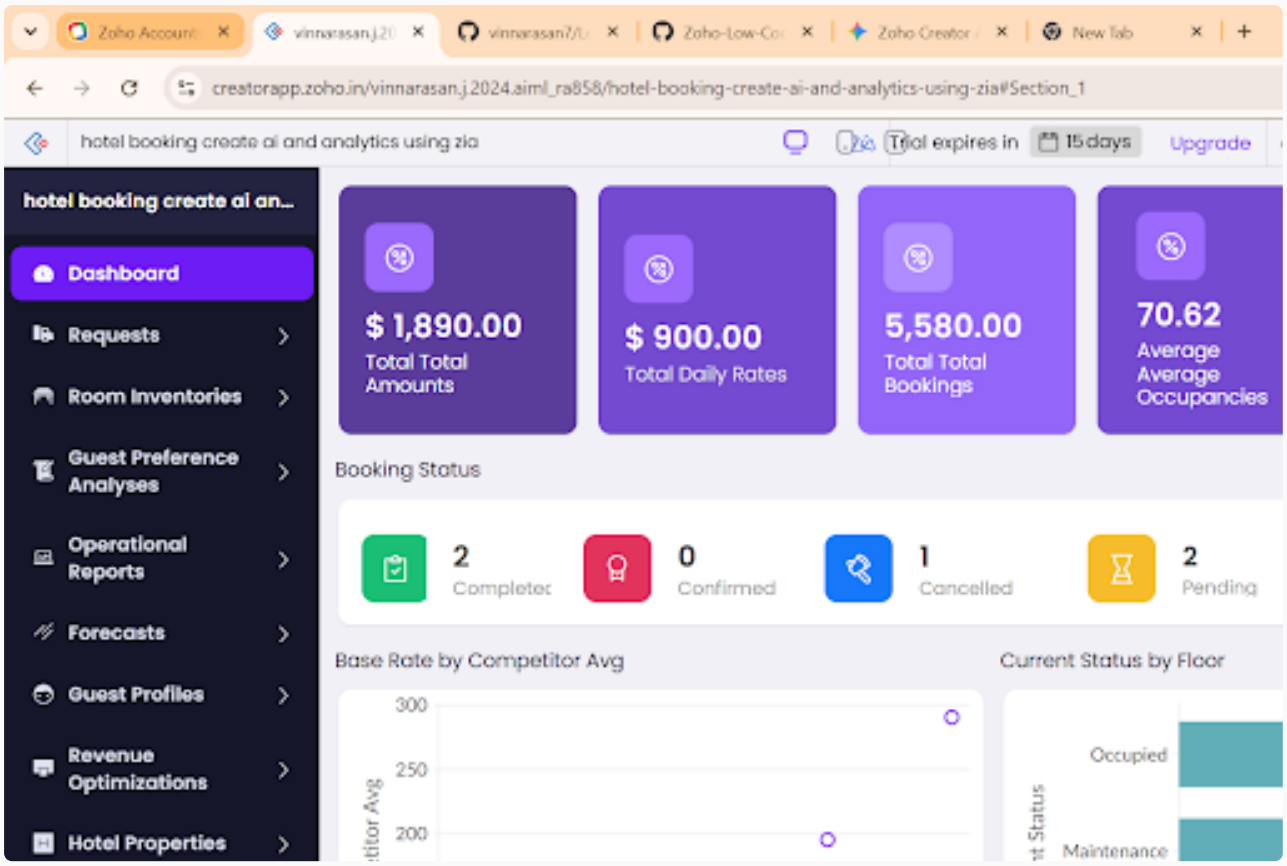


Figure 1: Main System Dashboard displaying real-time financial metrics, booking status, competitor rate analysis, and occupancy breakdown.

